Annotated Bibliography

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[3] The article provides a summary of techniques presently used in sustainable development within the scientific community. The study begins by discussing the importance of sustainable development, the frequency of various techniques used to analyze sustainable development, and important issues in sustainable transportation, such as logistics and environmental issues. It then dives into the techniques that are currently being used to analyze sustainable development, which includes simulation, optimization, machine learning, and fuzzy set analysis. The Optimization section covered the history of optimization used in transportation and the variety of applications for optimization, including in the difference between urban transportation systems and in global/regional transportation systems. Applications in simulation focus on the growing integration of simulation and optimization models and on usages in logistics, traffic congestion, and on the usage of demand-based transportation systems (ride sharing applications). The machine learning section primarily summarizes a paper that uses a random forest model to understand patterns in a bike sharing transportation system, as well as touches on a variety of techniques (random forest, logistic regression, neural networks, support vector machines, decision trees, heuristics, multi-agent systems, and multi-variate analysis) for a variety of analytical, traffic-related outcomes. The section on fuzzy sets takes note of the wide range of uses that recent projects have undertaken utilizing fuzzy sets. The paper notes that sustainability factors into many of the research projects, where businesses seek sustainability to increase innovation and enhance competitiveness. The paper concludes by suggesting that more of the investigated technologies be applied in tandem with one another to increase effectiveness.

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References

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[4] Michael Dorosan, Damian Dailisan, Jesus Felix Valenzuela, and Christopher Monterola. Use of machine learning in understanding transport dynamics of land use and public transportation in a developing city. *Cities*, 144:104587, 2024.